

Tirth Alkesh Patel

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EDUCATION

UCLA ANDERSON SCHOOL OF MANAGEMENT

Los Angeles, CA

Master of Science in Business Analytics (MSBA)

Sep 2025 – Dec 2026

- Machine Learning, Statistical Foundations of Analytics, Data Engineering, Operations Analytics, Optimization

INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

Chennai, India

Bachelor of Technology, Chemical Engineering

Dec 2021 – May 2025

- Mathematical Foundations of Data Science, Advanced Operations Research, Operations Management, Economics

SKILLS

Languages & Tools: Python, SQL, R, Git/GitHub, Docker, FastAPI, MySQL, PostgreSQL, Tableau

Machine Learning & Deep Learning: PyTorch, Transformers, Deep RL, XGBoost, Prophet, scikit-learn, Time Series Forecasting

NLP & GenAI: LangGraph, Agentic RAG, ChromaDB, LangChain, Claude & OpenAI APIs, VADER/NLTK, Embeddings

MLOps & Cloud: Databricks, Delta Lake, PySpark, Airflow, Snowflake, Model Monitoring, CI/CD (GitHub Actions), AWS EC2

Statistics & Experimentation: A/B Testing, Causal Inference, Regression Discontinuity, Hypothesis Testing, Feature Engineering

PROFESSIONAL EXPERIENCE

SEAT OF JOY – Stealth Startup

Chennai, India

Business & Strategy Analyst

Aug 2023 – Feb 2024

- Engineered a joint probability model over 20M+ rows in 100+ Census tables, fusing birth order and age gap distributions with cohort regression to size a 55M customer market, 45% more accurate than prior estimates.
- Led 3 analysts translating the sizing, pricing, and distribution models into a single investor narrative for non technical stakeholders, tracing every claim back to a model across 2 Shark Tank India rounds.
- Formulated a profit maximizing distribution model in Gurobi over state level shipping routes, encoding cost structure, demand caps from the sizing model, and a concentration penalty on any single state.

Business & Strategy Analyst Intern

Jun 2023 – Jul 2023

- Benchmarked 10+ competitors across 3 categories, sourcing production costs by cold call to rebuild competitor margins from first principles and set a defensible pricing band.

PROJECTS

Airline Crew Pairing: Transformer Deep Reinforcement Learning Scheduler

Deep RL, PyTorch, Neural Networks

- Designed a four layer ViT style transformer encoder with a learned context token and dueling pointer head to score 30 feature candidate tokens, trained by Double DQN with prioritized replay, Huber loss, and mixed precision on CUDA.
- Framed crew pairing as a sequential construction MDP over a 31 day, 1,013 leg network, encoding 9 duty and rest rules into a learned action mask so schedules stay feasible by construction.
- The trained policy generalized to a held out schedule with zero feasibility violations and 3.6x fewer fragile connections than the greedy construction baseline.

Wait Time Forecasting: Production ML Pipeline

Tableau, Databricks, PySpark, Delta Lake, Prophet, XGBoost

- Built an automated retraining pipeline on Databricks that retrains, validates, and promotes models weekly behind 12 quality gates over 840K+ records in a medallion Delta Lake, rolling back to the last validated forecast on failure.
- Engineered cyclical time and local wall clock features, then benchmarked Prophet and XGBoost fleets on a chronological 80/20 split of a 78K row holdout, cutting MAE 26.3% and severe misses 42.1% against a per ride mean baseline.
- Served 7-day forecasts across 109 attractions at 30-minute resolution to a downstream routing service, and built a live dashboard that publishes each run automatically at \$0/month infrastructure cost.

Forecast Driven Route Planner

Gurobi, Mixed Integer Programming, FastAPI

- Coupled a Prophet wait time forecast to an exact Gurobi MILP behind a FastAPI service, cutting query response time by 83.3% and token cost \$0.75 per query while saving guests 22 min of queue time.

Airbnb Superhost Badge: Causal Impact Study

Regression Discontinuity, VADER NLP

- Designed a quasi-experimental regression discontinuity study measuring the causal effect of Airbnb's Superhost badge on 6 KPIs at the 4.8 rating cutoff, separating demand-side engagement from guest-experience and perceived-value outcomes.
- Built a 125,685-host panel from ~3 GB across 34 U.S. metros, scoring reviews with VADER NLP and weighting ratings by review volume; HC1-robust OLS on 6,311 hosts isolated a significant jump in review volume alone ($p = 0.02$).

Course Material Q&A Assistant: RAG Chatbot

LangGraph, ChromaDB, FastAPI, LangChain, Docker

- Built a 13-node LangGraph RAG assistant over 4 course corpora where an LLM classifier routes 5 query types to the right tool; self-verifying deadline extraction re-queries ChromaDB and flags date conflicts with a confidence score.
- Deployed to production on Oracle Cloud using Docker and Caddy at ~\$4/month, with human-in-the-loop approval gating every embedding write and dual-LLM plus Vision fallbacks holding the service live through 9 production incidents.